MCS-104	DISTRIBUTED SYSTEMS	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To introduce fundamental principles of distributed systems, technical challenges and key design issues.
CO2	To impart knowledge of the distributed computing models, algorithms and the design of distributed system
CO3	To understand the concept of deadlock
CO4	To understand the Mechanism for building distributed file systems
CO5	To understand concept of Distributed Mutual Exclusion

MODULE I

Distributed System Concepts, Architectures, transparency Self management in Distributed system ,Thread, Virtualization, Client, server, code migration Semantics, Remote Procedure Calls, Communication, Naming, File System: Flat naming, Structure naming and Attribute based naming, Security, Concurrency control and recovery, local area network, distributed languages and communication primitives, case studies of distributed systems.

MODULE II

Clocks and Election algorithm, Consistency Model, Consistency Protocol, Resilience, Reliable communication, Distributed Commit, recovery in Distributed systems, security in distributed systems. Deadlock in distributed systems.

MODULE III

Distributed Operating Systems, Distributed File System, Sun NFS, and the Coda files system.NTFS, UNIX ext2 and ext3. Case studies of Distributed object based systems (CORBA) Distributed web based Systems.

References:

1. *P. K. Sinha, "Distributed Operating Systems," PHI.*
2. *Tanenbaum, A. S. and Van Steen, M. "Distributed Systems Principles and Paradigms, " (ISBN 0-13-088893-1), Prentice Hall 2002.*
3. *Bacon, J., "Concurrent Systems", 2nd Edition, (ISBN 0-201-177-676), Addison Wesley 1998.*
4. *Silberschatz, A., Galvin, P. and Gagne, G., "Applied Operating Systems Concepts", 1st Edition," (ISBN 0-471-36508-4), Wiley 2000..*
5. *Coulouris, G. et al, "Distributed Systems: Concepts and Design, 3rd Edition," (ISBN 0-201-61918-0), Addison Wesley 2001.*

Course Outcomes: After the completion of the course the student will be able to:

CO1	Illustrate the mechanisms of inter process communication in distributed system
CO2	Compare the concurrency control mechanisms in distributed transactional environment
CO3	Outline the need for mutual exclusion and election algorithms in distributed systems
CO4	Can apply the concept of distributed transaction
CO5	Can apply the concept of Dynamic voting protocols